REAL-WORLD DATA PROJECT

Retail Sales Data Analysis

Domain: Retail | **Project Type:** End-to-End Data Analysis

Prepared as an applied data science project demonstrating dataset preparation, exploratory analysis, visualization, and business insights.

Submitted By: _____

Course / Class: _____

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1. Project Overview

This project analyzes retail transaction data to understand sales performance across product categories, regions, and time. The workflow follows a practical data science pipeline: dataset creation and preparation, data cleaning, exploratory data analysis, visualization, interpretation, and conclusion.

Objectives

• Measure overall sales, units sold, and profit.
• Compare performance across product categories and regions.
• Identify monthly sales trends.
• Convert the analysis into actionable business observations.

2. Dataset Description

For this educational project, a realistic retail transaction dataset was prepared with 240 transaction records. Each record contains date, region, product category, units sold, unit price, sales, profit margin, and profit. The dataset is suitable for demonstrating an end-to-end real-world analytics workflow.

Column	Description
Date	Transaction date
Region	Sales region: North, South, East, or West
Category	Product category
Units_Sold	Number of units sold
Unit_Price	Price per unit
Sales	Units sold × unit price
Profit_Margin	Estimated profit percentage
Profit	Estimated profit generated

3. Data Preparation and Cleaning

The dataset was checked for missing values, duplicate records, invalid numeric values, and inconsistent category labels. Dates were converted to a standard date format, and Sales and Profit were treated as numeric measures. Derived measures were calculated from units sold, price, and profit margin. These steps make the dataset suitable for analysis and visualization.

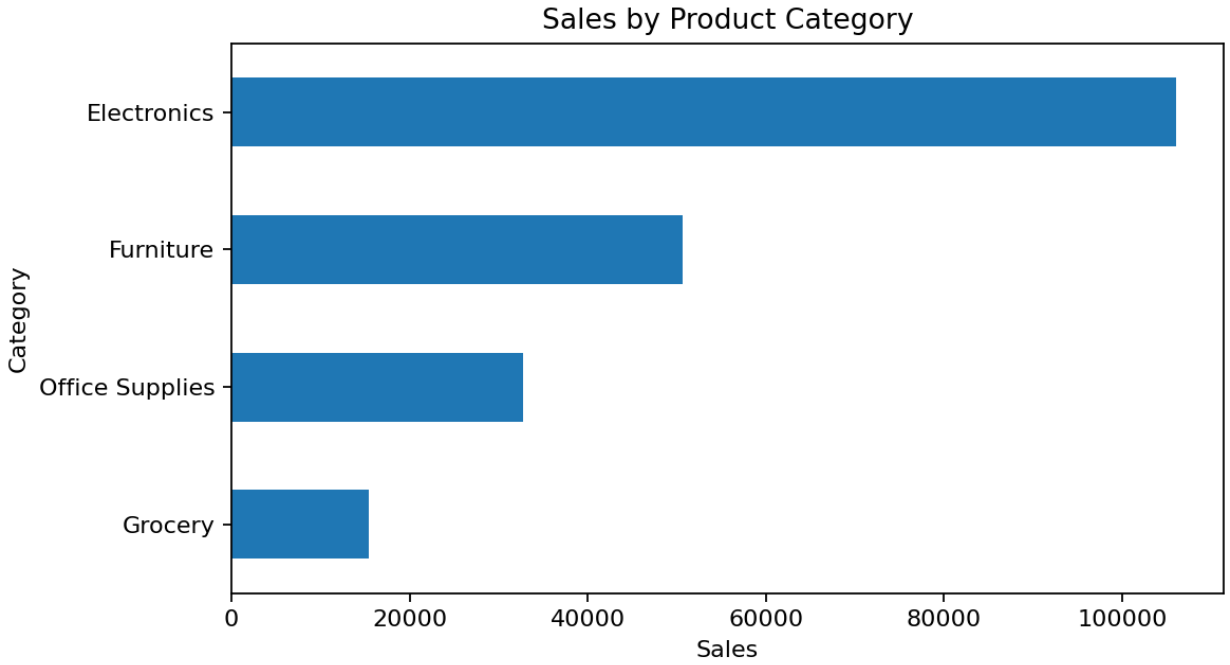
4. Key Performance Indicators

Metric	Result
Total Sales	$204,920.30
Total Profit	$33,256.15
Total Units Sold	1,792
Average Transaction Sales	$853.83

5. Exploratory Data Analysis

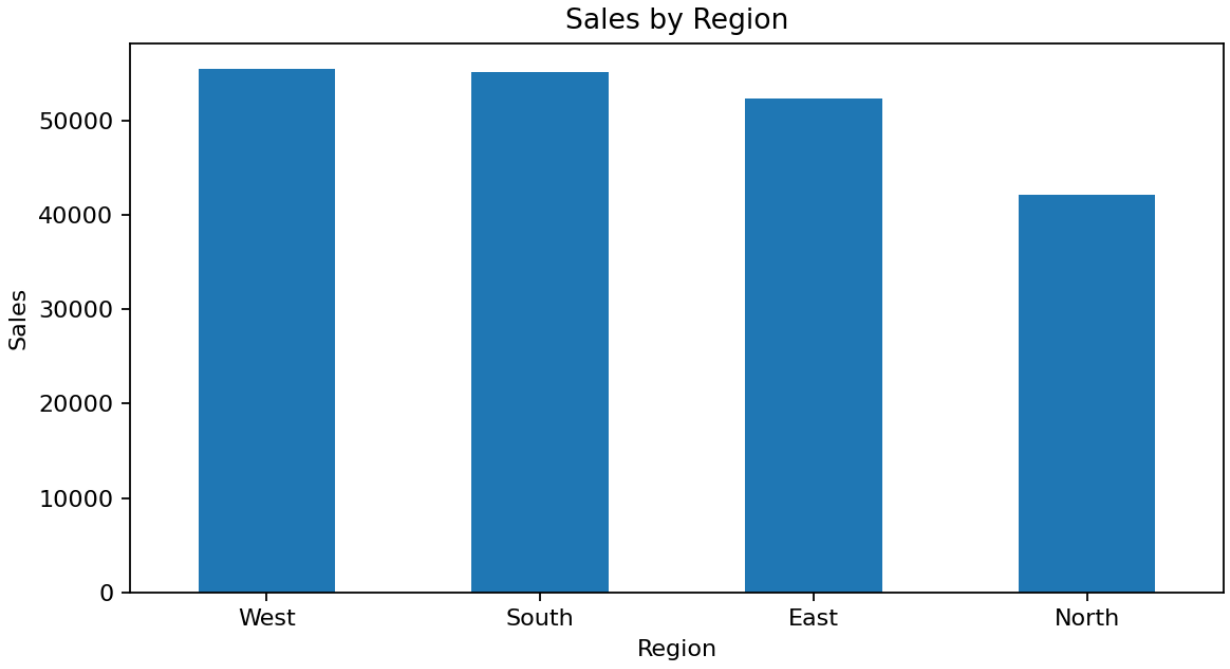
Sales by Product Category

The category-level analysis shows that **Electronics** generated the highest sales, while **Grocery** generated the lowest sales among the four categories in this sample.



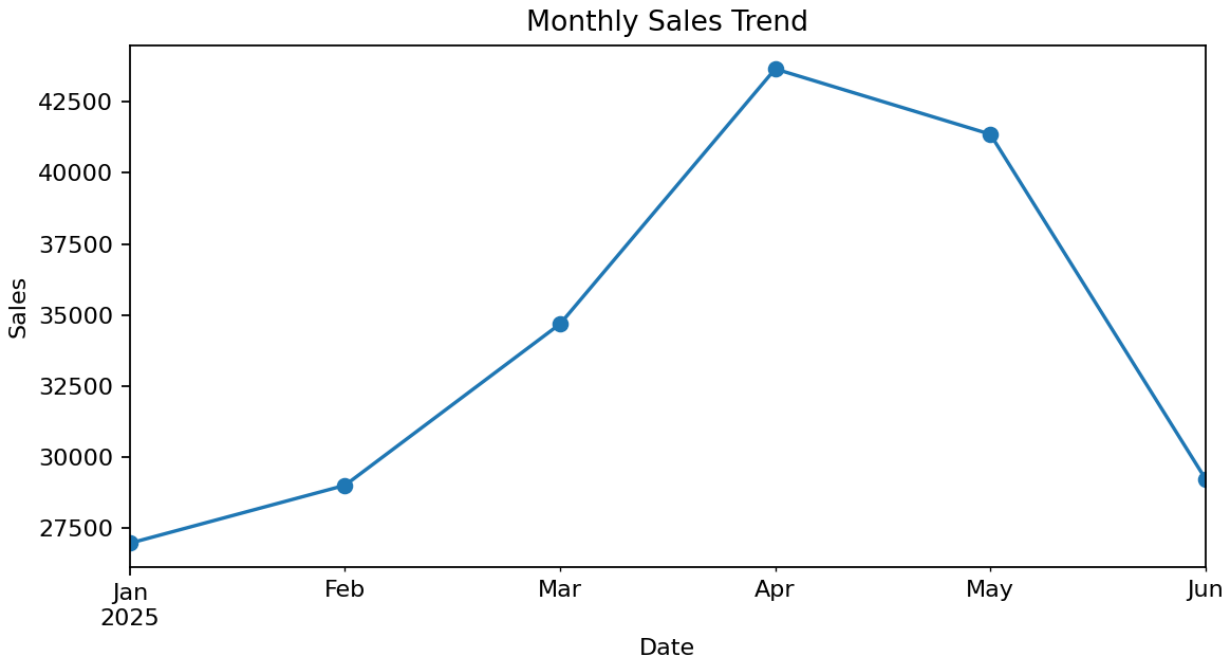
Sales by Region

The **West** region recorded the highest total sales in the dataset. Regional comparison can help management identify where demand is strongest and where additional sales activity may be needed.



6. Time-Series Analysis

Monthly aggregation was used to observe changes in sales over time. The line chart below helps identify periods of relatively high or low sales and can support inventory planning, promotions, and staffing decisions.



7. Findings

1. Total sales across the sample were **$204,920.30**, with **1,792** units sold.
2. **Electronics** was the largest sales category in this dataset.
3. **West** was the highest-sales region in the sample.
4. Sales vary over time, making monthly monitoring useful for operational planning.
5. Profit should be considered alongside sales because high revenue does not automatically imply the highest profitability.

8. Business Recommendations

• Monitor the highest-performing categories to maintain adequate inventory.
• Compare regional performance regularly and investigate large performance gaps.
• Use monthly sales trends to support inventory and promotional planning.
• Track both revenue and profit margin when evaluating products.
• Extend the analysis with customer-level data, discounts, returns, and marketing spend for deeper prediction.

9. Conclusion

This project demonstrates how a retail dataset can be transformed into useful business information through an end-to-end data analysis workflow. Data preparation, descriptive statistics, category and regional comparisons, and time-series visualization provide a foundation for evidence-based retail decision-making. The same workflow can be extended to predictive modeling, such as sales forecasting or customer purchase prediction, when a larger historical dataset is available.

10. Tools and Techniques Used

Python: Data processing and analysis
Pandas / NumPy: Data preparation and numerical analysis
Matplotlib: Data visualization
ReportLab: Project report generation
Methods: Data cleaning, aggregation, descriptive analysis, KPI calculation, and visualization

— End of Project Report —